R18

Code No: 152AN

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech I Year II Semester Examinations, July/August - 2021
CHEMISTRY
(Common to EEE, CSE)

Time: 3 Hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) Draw the molecular orbital diagram of Oxygen molecule. Calculate the bond order.
b) Explain the postulates of molecular orbital theory. [7+8]

2.a) Explain the salient features of crystal field theory.
b) Explain the crystal field splitting of d-orbitals in square planar geometries. [7+8]

3.a) A sample of water on analysis contains 8.4 mg/l of Magnesium bicarbonate, 24.0 mg/l of Magnesium sulphate, 8.1 mg/l of Calcium bicarbonate, 11 mg/l of Calcium chloride and 13.6 mg/l of Calcium sulphate. Calculate the total, permanent and temporary hardness of the sample and also express the result in degree Clark and degree French.
b) Discuss the estimation of hardness of water by complexometric method using EDTA. [7+8]

4.a) Discuss colloidal and Calgon conditioning processes.
b) What are various boiler troubles? Discuss the caustic embrittlement. [7+8]

5.a) Describe the construction and reactions involved in charging and discharging of Pb- Acid storage battery.
b) Calculate the EMF of the following cell.
$Zn/Zn^{+2}(0.025M)//Fe^{+2}(0.015M)/Fe$
(E^0 of Zn^{+2}/Zn = -0.763 v and Fe^{+2}/Fe = 0.44 V). [7+8]

6.a) What is Electro chemical series? What are the conclusions from the series? Explain.
b) Discuss the mechanism of Electro chemical corrosion. [7+8]

7.a) Write the mechanism involved in the oxidation of alcohols using $KMnO_4$.
b) Explain the synthesis and applications of paracetamol. [7+8]

8.a) Discuss the different types of possible excitations due to UV- visible absorption.
b) Explain the principle involved in NMR spectroscopy. [7+8]

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